

~~18.5~~
~~20~~

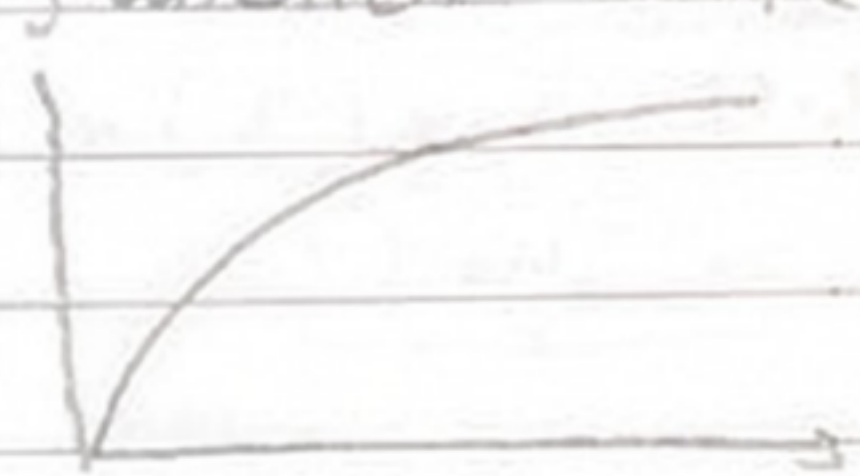
→ Malak sameh Abdelnazi

22013 (97)

- (Q1) a) intensity values: dark due to writing $r < 100$ dominant $r > 200$ ✓
 grey color exists due to shading $100 \leq r < 170$ ✓, white color of the paper
 b) To increase contrast use "contrast stretching"

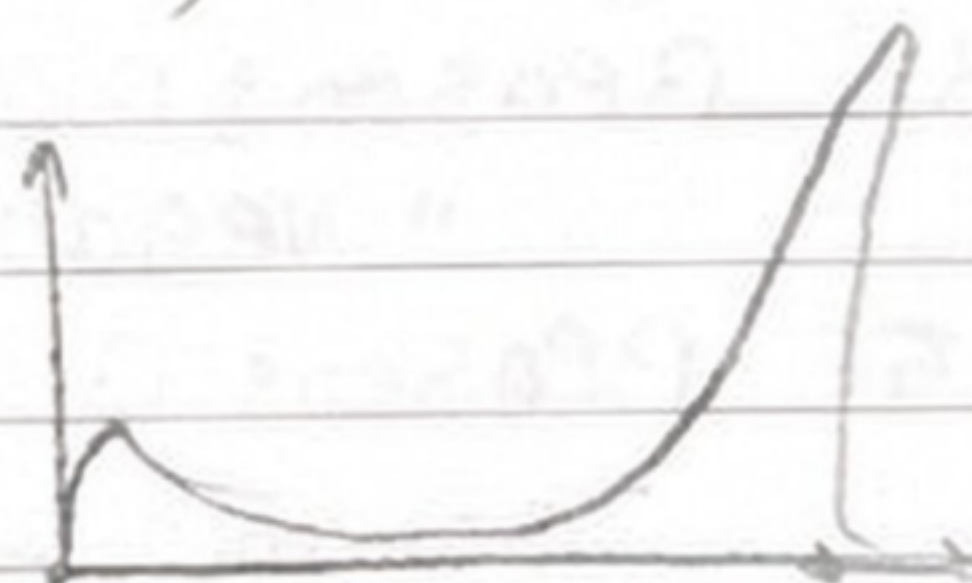


- To remove shading use "Gamma function" while $r < 1$
 for back ground



- To make text clearer use sharpening filter as (high boost) or laplace

c) expected output:



good output image enhanced
 not binarized - 0.5

~~3.5~~
~~4~~

(Q2) A(3,3) B(7,3) C(7,7) D(3,7)
 A'(6.5,3.4) B'(12.5,3.4) C'(12.5,6.6) D'(6.5,6.6)

$$x' = S_x x + T_x$$

$$y' = S_y y + T_y$$

$$W = \frac{6}{4} = 1.5$$

$$L = \frac{6 \cdot 6 - 3 \cdot 4}{7 - 3} = 0.8$$

$$S_x = 1.5, S_y = 0.8$$

$$\rightarrow 6.5 = (1.5)(3) + T_x$$

$$2 T_x = 2$$

$$\rightarrow 3.4 = 0.8(3) + T_y$$

$$2 T_y = 1$$

$$\begin{bmatrix} 1.5 & 0 & 2 \\ 0 & 0.8 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

(Q3) GE - TB - H3 ✓ Reason: Black turned to white a vice versa
 "Negative"

GF - TE - H5 ✓ Reason: Black and white image only
 so only 2 values (Binary)

GR - TC - H4 ✓ Reason: image got darker so histogram
 is shifted to the left

GC - TD - H1 ✓ Reason: image is brighter, histogram
 should be right shifted

GD - TA - H2 ✓ Reason: image is slightly brighter
 than the original one, histogram
 slightly right shifted

→ Graph A not used "it's for original graph"

Excellent

8/8

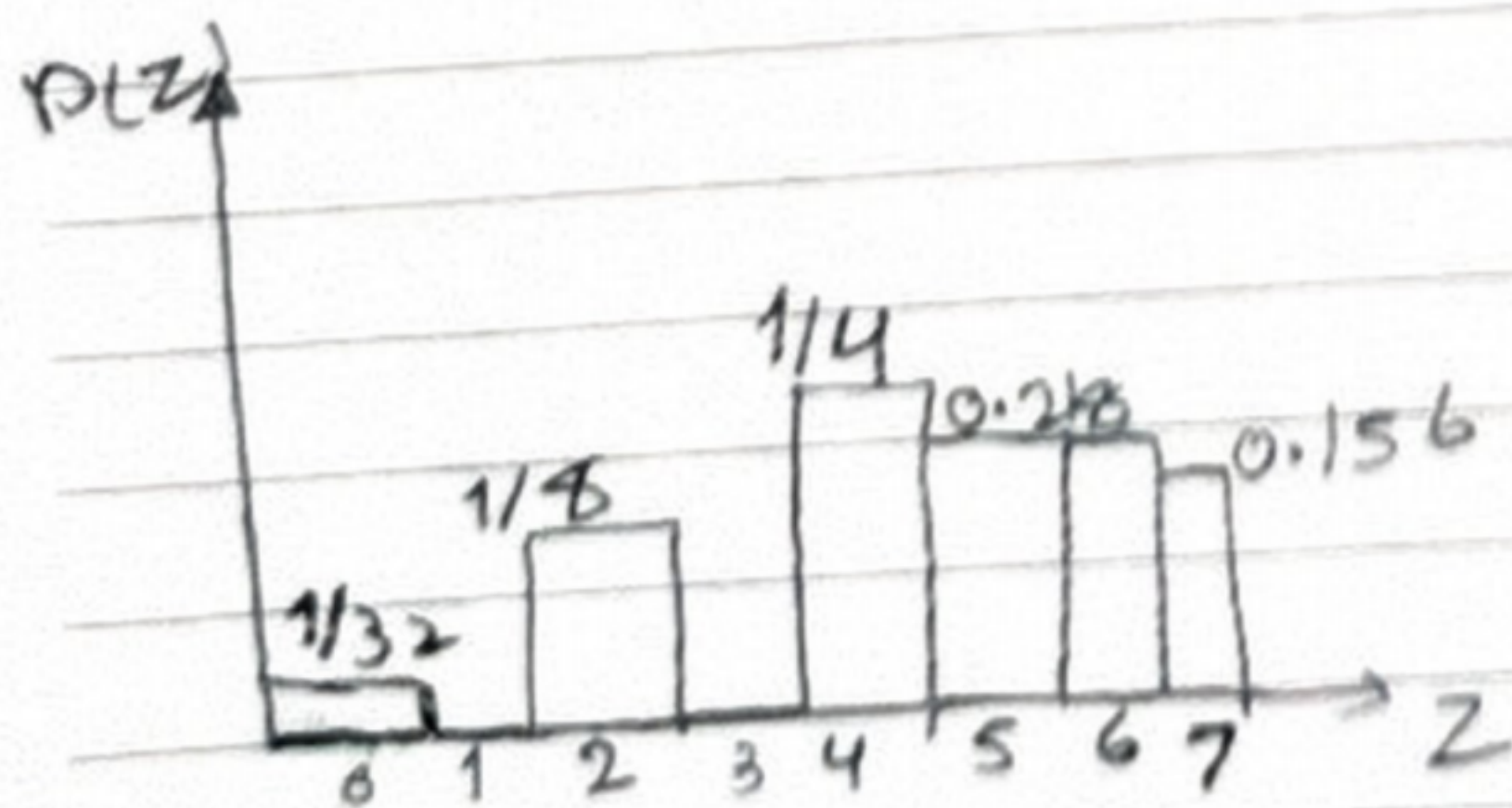
(Q4)

r	n_r	CDF
0	1	$1/32$
1	4	$5/32$
2	8	$13/32$
3	5	$9/16$
4	2	$5/8$
5	2	$11/16$
6	5	$27/32$
7	5	1

z	n_z	CDF
0	0	0
1	0	0
2	4	$1/8$
3	4	$1/4$
4	4	$3/8$
5	4	$1/2$
6	5	$21/32$
7	11	1

r	s	$G(z) \rightarrow Y$
0	0	0
1	1	0
2	3	1
3	4	2
4	4	3
5	5	4
6	6	5
7	7	7

r	z
0	0
1	2
2	4
3	5
4	5
5	6
6	6
7	7



Where is mapped image?

-1

didn't ask for plot

Revise well!